



Read Two Scripts Carefully

Quick facilitation notes & answer key

Goal

The first sheet of the debugging subject builds the foundation skill: reading two concurrent scripts accurately. BOTH scripts here are correct (no bug yet) — students trace each one move by move and confirm the combined outcome. Accurate reading of code that involves concurrent events is the Grade-2 C3.2 skill every later debugging sheet depends on.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters (green Bit, purple Pixel) on a 0-5 path; trace each script one move at a time before putting the results together.

Run it (about 20 min)

1. I DO: trace Bit out loud — 'home 0, forward 2 → 2, forward 1 → 3.' Point to each block as you read it.
2. WE DO: trace Pixel together — 'home 5, back 1 → 4, back 1 → 3.'
3. YOU DO: students record both landings and notice both are on 3 (they meet).
4. Connect: 'Read each script carefully on its own, then put them together — that is how we will find bugs next.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 0 → forward 2 → 2 → forward 1 → 3. Bit lands on 3.

Ex 2 — Pixel: 5 → back 1 → 4 → back 1 → 3. Pixel lands on 3.

Ex 3 — both start on the one green flag; Bit lands on 3 and Pixel lands on 3 → yes, the same number (they meet).

Reading note: there is no bug on this sheet — the point is accurate tracing, which the next sheets build on to find bugs.

Watch for & differentiate

Common error: dropping the second move — remind students each script has TWO moves to read in order.

Common error: starting both from the same home — Bit starts at 0, Pixel at 5; read each hat's sprite.

Simplify: trace one script fully, then the other, before combining.

Extend: ask students to add one more move to either script that would still land both on the same number.

Success indicator: the child lands both on 3 by reading each script accurately, move by move.